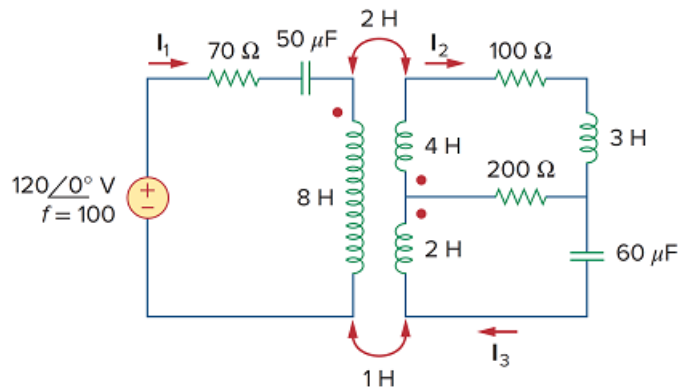


Problem

Use *PSpice* or *MultiSim* to find I_1 , I_2 , and I_3 in the circuit of Fig. 13.142.

Figure 13.142



Step-by-step solution

Step 1 of 2

The schematic is shown below.

$$k_1 = \frac{2}{\sqrt{4 \times 8}} = 0.3535 ,$$

$$k_2 = \frac{1}{\sqrt{2 \times 8}} = 0.25$$

In the AC sweep box, we let total pts = 1, start frequency = 100, end frequency = 100. After simulation, the output file includes,

Frequency	1 M (V_Print1)	1 P (V_Print1)
1.000 E+02	1.0448 E-01	1.396 E+01
Frequency	1 M (V_Print2)	1 P (V_Print2)
1.000 E+02	2.954 E-02	-1.438 E+02
Frequency	1 M (V_Print1)	1 P (V_Print1)
1.000 E+02	2.088 E-01	2.440 E+01

i.e.,

$$I_1 = 104.5 \angle 13.96^\circ \text{ mA}$$

$$I_2 = 29.54 \angle -143.8^\circ \text{ mA}$$

$$I_3 = 208.8 \angle 24.4^\circ \text{ mA} .$$

Step 2 of 2

